

GED Math Assignment 7

$$\begin{aligned} \text{A. 1. } P &= 10 + 7.8 + 13 \\ &= 30.8 \text{ units} \end{aligned}$$

$$\begin{aligned} A &= \frac{1}{2}bh \\ &= \frac{1}{2}(13)(6) \\ &= 39 \text{ units}^2 \end{aligned}$$

$$\begin{aligned} 2. P &= 4 + 4 + 4 + 4 \\ &= 16 \text{ units} \end{aligned}$$

$$\begin{aligned} A &= 4 \times 4 \\ &= 16 \text{ units}^2 \end{aligned}$$

$$\begin{aligned} 3. P &= 20 + 32 + 20 + 32 \\ &= 104 \text{ units} \end{aligned}$$

$$\begin{aligned} A &= 20 \times 32 \\ &= 640 \text{ units}^2 \end{aligned}$$

$$\begin{aligned} 4. P &= 4.3 + 8 + 4.3 + 8 \\ &= 24.6 \text{ units} \end{aligned}$$

$$\begin{aligned} A &= bh \\ &= 8 \times 3.7 \\ &= 29.6 \text{ units}^2 \end{aligned}$$

$$\begin{aligned} 5. P &= 14 + 24 + 42 + 29 \\ &= 109 \text{ units} \end{aligned}$$

$$\begin{aligned} A &= \frac{1}{2}(a+b)h \\ &= \frac{1}{2}(14 + 42)(22) \\ &= 616 \text{ units}^2 \end{aligned}$$

$$\begin{aligned} 6. P &= 12 + 4 + 12 + 4 \\ &= 32 \text{ units} \end{aligned}$$

$$\begin{aligned} A &= bh \\ &= (4)(9.5) \\ &= 38 \text{ units}^2 \end{aligned}$$

$$\begin{aligned} 7. \text{ Area of trapezoid} &= \frac{1}{2}(a+b)h \\ &= \frac{1}{2}(5+9)(8) \end{aligned}$$

→ sorry, "retangle"
should be "rectangle" = 56 in²

$$\begin{aligned} 8. A &= L \times W \\ &= 9 \times 6 \\ &= 54 \text{ ft}^2 \end{aligned}$$

$$\begin{aligned} 9. \text{ Area of Patio} &= 9 \times 12 \\ &= 108 \text{ ft}^2 \\ \text{Total Area} &= 18 \times 24 \\ &= 432 \text{ ft}^2 \text{ (over)} \end{aligned}$$

$$\begin{aligned}\text{Area of grass} &= 432 - 108 \\ &= 324 \text{ ft}^2\end{aligned}$$

$$\begin{aligned}10. \quad P &= 6(4) \\ &= 24 \text{ cm}\end{aligned}$$

PART 2

$$\begin{aligned}A \quad 1. \quad C &= 2\pi r \\ &= 2\pi(10) \\ &= 62.8 \text{ in}\end{aligned}$$

$$\begin{aligned}A &= \pi r^2 \\ &= \pi(10)^2 \\ &= 314.2 \text{ in}^2\end{aligned}$$

$$\begin{aligned}2. \quad C &= 2\pi(2) \\ &= 12.6 \text{ cm}\end{aligned}$$

$$\begin{aligned}A &= \pi(2)^2 \\ &= 12.6 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}3. \quad C &= 2\pi(4) \\ &= 25.1 \text{ m}\end{aligned}$$

$$\begin{aligned}A &= \pi(4)^2 \\ &= 50.3 \text{ m}^2\end{aligned}$$

$$\begin{aligned}4. \quad P &= 2\pi r \\ &= 2\pi(6) \\ &= 38 \text{ ft}\end{aligned}$$

$$\begin{aligned}7. \quad \text{Diameter of outer 10-point} \\ \text{Circle} &= 2 + 2 + 2 \\ &= 6 \text{ in.}\end{aligned}$$

Part 3

$$\begin{aligned}A \quad 1. \quad V &= L \times W \times h \\ &= 4 \times 5 \times 8 \\ &= 160 \text{ units}^3\end{aligned}$$

$$\begin{aligned}\text{So, } C &= 2\pi r \\ &= 2\pi(3) \\ &= 19 \text{ in.}\end{aligned}$$

$$\begin{aligned}2. \quad V &= 3 \times 3 \times 3 \\ &= 27 \text{ units}^2\end{aligned}$$

$$\begin{aligned}
 3. \quad V &= \pi r^2 h \\
 &= \pi (3)^2 (5) \\
 &= 141 \text{ units}^3
 \end{aligned}$$

$$\left(\begin{aligned} r &= \frac{d}{2} \\ &= \frac{6}{2} \\ &= 3 \end{aligned} \right)$$

$$\begin{aligned}
 4. \quad V &= L \times W \times h \\
 &= 10 \times 6 \times 7 \\
 &= 420 \text{ units}^3
 \end{aligned}$$

$$\begin{aligned}
 5. \quad V &= 1.5 \times 1.5 \times 1.5 \\
 &= (1.5)^3 \\
 &= 3.38 \\
 &= 3 \text{ units}^3
 \end{aligned}$$

$$\begin{aligned}
 6. \quad V &= \pi r^2 h \\
 &= \pi (5)^2 (3) \\
 &= 236 \text{ units}^3
 \end{aligned}$$

$$\begin{aligned}
 7. \quad V &= L \times W \times h \\
 80 &= 4 \times 4 \times h \\
 80 &= 16h \\
 \frac{80}{16} &= h
 \end{aligned}$$

$$5 = h$$

∴ The height of the box is 5 ft.

$$\begin{aligned}
 8. \quad V &= L \times W \times h \\
 &= 5 \times 5 \times 5 \\
 &= 125 \text{ ft}^3
 \end{aligned}$$

∴ The volume of the crate is 125 ft^3

$$\begin{aligned}
 9. \quad V &= \pi r^2 h \\
 &= \pi (6)^2 (3) \\
 &= 339.3 \text{ ft}^3
 \end{aligned}$$

$$\begin{aligned}
 \text{so, } \# \text{ Scoops} &= \frac{339.3}{20} \\
 &= 16.96
 \end{aligned}$$

∴ 17 scoops are needed for the pool.